

Iliac Crest Bone Graft Harvest for Alveolar Cleft Repair: A Systematic Review Comparing Minimally Invasive Trephine and Conventional Open Techniques

Collecte de greffon osseux de crête iliaque pour réparation de fente palatine : Revue systématique comparant une technique par tréphine peu invasive et les techniques conventionnelles ouvertes

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Abstract

Background: In the course of developing a standardized pathway for alveolar cleft repair, we conducted a systematic review comparing minimally invasive trephine with conventional open technique for iliac crest bone graft harvest in a pediatric population. **Methods:** A systematic review was conducted of studies comparing open with minimally invasive trephine techniques in pediatric populations undergoing alveolar cleft repair. Exclusion criteria included reviews, case series, editorials, abstracts, and those with adult-only populations. Data were compiled with outcome variables selected *a priori*. **Results:** Of 422 manuscripts screened, five met criteria. These comprised 257 patients (116 open, 141 trephine). Average age was 11 years. Patients undergoing trephine harvest had reduced length of stay (1.0-5.0 days trephine vs 1.25-5.4 days open), time to unassisted ambulation (16-46 hours vs 20-67 hours open), and less postoperative narcotic use (0.31 mg/kg vs 1.64 mg/kg IV morphine). Volume of cancellous bone was reported as 2.53 mL for open versus 1.22 mL for trephine in one study, and trephine graft was supplemented with demineralized bone in 54% of cases in another study. The use of anesthetic adjuncts was inconsistent but had a significant effect on postoperative pain and ambulation. **Conclusions:** Compared to open techniques, the minimally invasive trephine bone graft harvest is associated with a shorter time to discharge, slightly lower infection rates, and reduced opioid use. The possible benefits of trephine harvest must however be balanced against the risk of insufficient graft harvest. Finally, the choice of perioperative analgesic adjuncts significantly impacts patient outcomes regardless of the technique employed.

Résumé

Contexte: Dans le cours de l'élaboration d'une méthode standardisée de réparation des fentes palatines, nous avons réalisé une revue systématique comparant la technique par tréphine peu invasive à la technique conventionnelle ouverte pour la collecte d'os de la crête iliaque dans une population pédiatrique. **Méthodes:** Une revue systématique a été réalisée sur les études comparant les techniques ouvertes avec les techniques utilisant une tréphine peu invasive dans des populations pédiatriques subissant une réparation de fente palatine. Les critères d'exclusion incluaient les revues, les séries de cas, les éditoriaux, les résumés et les études ne portant que sur une population adulte. Les données ont été compilées avec des variables de résultats sélectionnées *a priori*. **Résultats:** Parmi les 422 manuscrits sélectionnés, cinq répondaient aux critères. Ils incluaient 257 patients (116 techniques ouvertes, 141 tréphines). L'âge moyen des patients était de 11 ans. Les patients pour lesquels une collecte avait

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été obtenue par tréphine avaient une durée de séjour plus courte (tréphine : 1,0 à 5,0 jours; contre technique ouverte : 1,25 à 5,4 jours), un délai de déambulation non aidée plus court (16 à 46 heures contre 20 à 67 heures pour la technique ouverte) et un moindre recours aux antalgiques postopératoires (0,31 mg/kg contre 1,64 mg/kg de morphine IV). Le volume d'os spongieux collecté était de 2,53 ml avec les techniques ouvertes contre 1,22 ml avec la tréphine; le greffon par tréphine a été complété par de l'os déminéralisé dans 54% des cas dans une autre étude. L'utilisation de suppléments anesthésiques n'était pas homogène, mais a eu un effet significatif sur la douleur postopératoire et la déambulation. **Conclusions:** Comparativement aux techniques ouvertes, la collecte peu invasive de greffon osseux par tréphine est associée à une durée de congé plus courte, des taux d'infections légèrement inférieurs et à une moindre utilisation d'opioïdes. Toutefois, les avantages éventuels de la collecte par tréphine doivent être soupesés par rapport au risque de collecte insuffisante de greffon. Enfin, le choix de suppléments analgésiques périopératoires a des répercussions significatives sur l'évolution des patients, indépendamment de la technique employée.

Keywords

analgesics, opioid, bone graft, cleft palate, ilium, length of stay

Introduction

Autologous iliac crest bone grafting (ICBG) is the gold standard for pediatric alveolar cleft repair.^{1,2} The most common techniques for pediatric ICBG harvest are the conventional open technique and minimally invasive trephine harvest.^{3–5} Despite reports of surgeon and institutional preferences, evidence-based consensus on the outcomes of these techniques does not exist.

Trephine harvest of ICBG in the adult population has been well studied,⁴ and several pediatric centres have adopted this technique. Reported benefits include shorter surgical time with a smaller incision, shorter time to ambulation and discharge, and decreased postoperative opioid use.^{3–5} Within a pediatric group, it is unclear whether this technique achieves sufficient volume of cancellous bone, if there is a notable risk of visceral organ damage, and whether the donor morbidity is comparable to open techniques.^{1,6–8}

Our institution has recently developed a novel clinical care pathway for pediatric alveolar cleft repair. As part of this process, we sought to identify the optimal surgical approach for ICBG harvesting. A systematic review was conducted, comparing the use of trephine to conventional open ICBG harvest in pediatric patients undergoing alveolar cleft repair.

Methodology

A systematic review was conducted as per Preferred Reporting Items for Systematic Review and Meta-Analyses principles (PRISMA) specifications.⁹ MEDLINE, EMBASE, and Scopus databases were searched on January 24, 2021. Manuscripts were included if they compared open to trephine ICBG harvest in a pediatric population (Defined as a mean study age under 18), undergoing alveolar cleft repair. Exclusion criteria were review articles, case series, commentaries, editorials, abstracts, and articles with a focus on adult populations. Search terms were created with the assistance of a medical librarian (H.L.R.). (Full MEDLINE search terms in Appendix 1). Two investigators independently reviewed all citations for selection for full-text review (A.R.T. and S.F.).

Data were organized based on outcome variables selected *a priori*, including patient demographics, operative time, time to unassisted ambulation, length of hospital stay, postoperative pain control and pain scores, volume of bone harvested, use of bone supplements, and complications. Quantitative results were not amenable to meta-analysis due to heterogeneity in outcome reporting. As such, ranges of the selected outcome values were reported based on the minimum and maximum of the means reported in each study. Analysis was organized into three themes: (i) donor site morbidity; (ii) volume and type of harvested bone; and (iii) safety.

Results

Of 422 manuscripts initially screened, five studies were ultimately included in the analysis (Figure 1: PRISMA flow diagram).^{5,10–13} Reasons for exclusion of full-text reviewed articles were: Non-ABG patient population (3), Case Study (1), Technical descriptor/no comparison (6), Alternative minimally invasive technique (3), Alternate grafting technique (1). Of the papers reviewed, they included 257 patients; 116 patients underwent conventional open harvest and 141 patients underwent trephine harvest. The average of the mean reported ages was 11.5 years in the trephine group (range 6.5–30 years), and 11.1 years in the open (range 6–25 years). The open technique included 74 patients with unilateral clefts and 45 patients with bilateral clefts (total 164 sites for grafting). Trephine groups included 88 patients with unilateral clefts, and 56 patients with bilateral clefts for a total of 200 sites (Table 1).

Operative Time

Patients undergoing trephine harvest had a reduced length of stay (range: 1.0–5.0 days vs 1.25–5.4 days for open harvest), and shorter time to unassisted ambulation (range: 16–46 hours vs 20–67 hours for open harvest) (Table 1). Total operative time was shorter in the trephine harvest group (1.9–2.4 hours vs 2.5–3.3 hours).^{5,10,11} One study reported the operative time

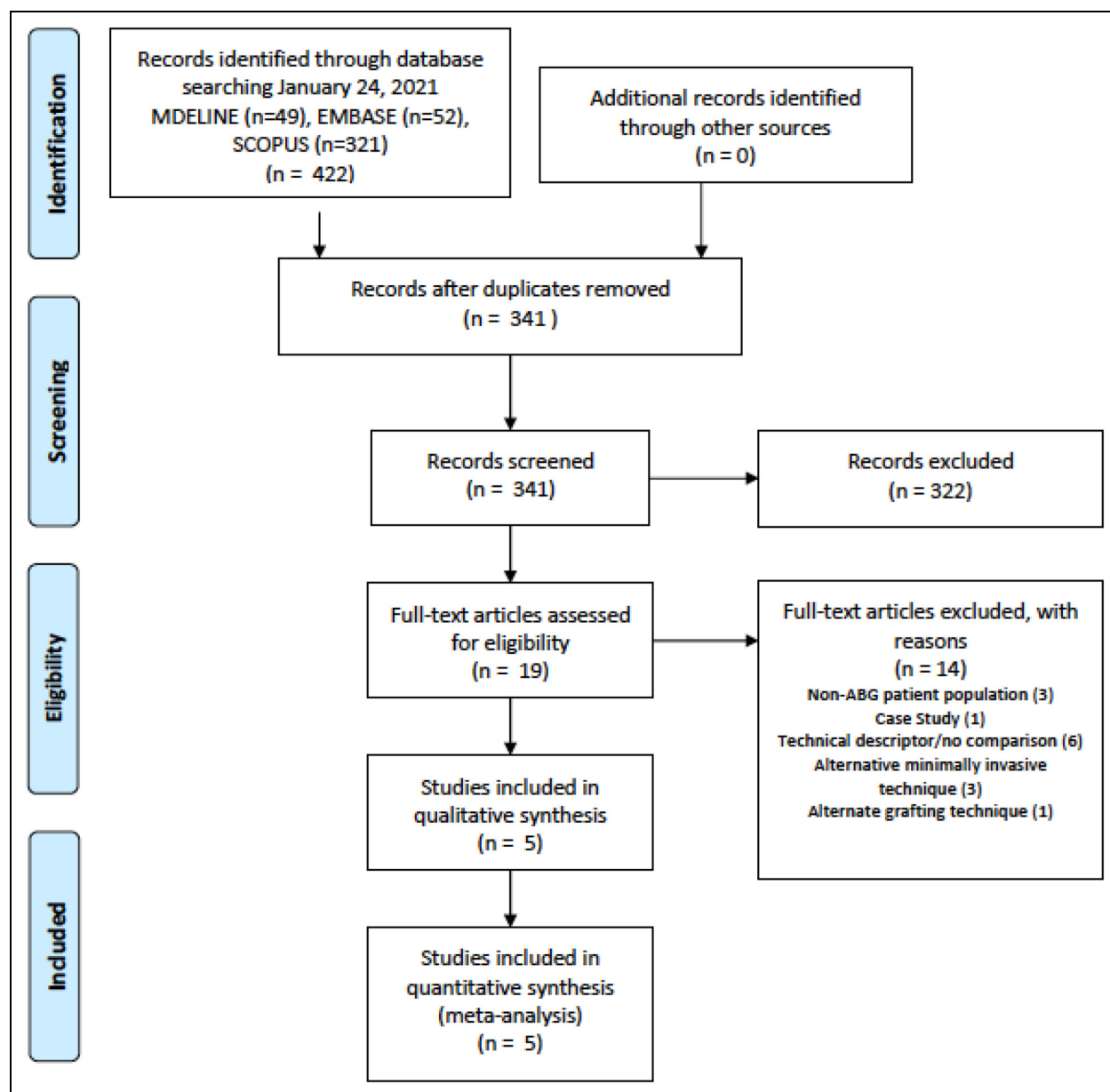


Figure 1. PRISMA flow sheet.

required for the harvest portion only, reporting 11.4 minutes for trephine harvest versus 20 minutes for open harvest.¹²

Pain Control

Peri- and postoperative pain control varied widely between studies. In three studies, there was no reported use of pain adjuncts, such as local anesthetic or regional blocks.^{5,11,13} In the study by Bykowski et al, the analgesic approach varied between three study groups; the open group received no pain adjuncts, while the “trephine-only” group had GelFoam® collagen sponge (Pfizer) impregnated with bupivacaine placed at the donor site.¹⁰ They additionally compared these to a trephine harvest with pain pump group, in which patients received an

infraorbital nerve block, IV acetaminophen infusion, GelFoam® at the donor site, and a catheter infusion of ropivacaine.¹⁰ They found that the addition of an anesthetic infusion at the donor site significantly reduced the length of hospital stay and pain scores on both open and trephine harvest groups (length of stay for open 2.4 days, vs trephine 2.7 days, vs trephine plus pain pump 0.5 days, vs no harvest control 0.9 days).¹⁰ Intraoperatively, Burstein et al used local infiltration with bupivacaine subcutaneously and in the deep periosteal plane, as well as placed GelFoam® impregnated with bupivacaine at the donor site.¹² Postoperative analgesic ranged from the use of acetaminophen and codeine most commonly,^{5,10,12} to the use of patient-controlled analgesia (PCA) as needed,⁵ or acetaminophen and meperidine¹¹ (Table 2).

Table 1. Open Versus Trephine Iliac Crest Bone Graft Harvest.

	Open ICBG (n = 116)	Trephine ICBG (n = 141)
Age ^a (years)	11.1 (6-25)	11.5 (6-30)
Unilateral cleft patients	74	88
Bilateral cleft patients	45	56
Total cleft sites	164	200
Postoperative length of stay (days) ^b	1.3-5.4	1.0-5.0
Time to unassisted ambulation (hours)	20-67	16-46
Operative time (hours)	2.5-3.3	1.9-2.4

Abbreviation: ICBG, iliac crest bone graft.

^aAge: Reported as an average of the means reported per study; heterogeneity of data precluded meta-analysis of values.

^bTrephine PLUS pain pump had length of stay 0.5 days.¹⁰

Postoperative narcotic consumption and pain scores were reported in two studies.^{5,10} One study found reduced narcotic use post operatively in the trephine group (0.31 vs 1.64 mg/kg IV morphine).⁵ They also reported higher pain scores for open harvest at time of discharge (3.1/10 vs 0.26/10).⁵ Another study reported shorter duration of postoperative analgesia in the form of acetaminophen and codeine in the trephine versus open groups (16.5 hours vs 26 hours).¹⁰

Bone Graft Harvest Volume

The volume of cancellous bone harvest was reported in two studies. The volume harvested in one study was 2.53 mL in the open group and 1.22 mL in trephine group. All patients undergoing trephine harvest required supplementation with deproteinized bovine bone (DBB).¹¹ Another study supplemented with demineralized bone graft in 54% of patients who underwent trephine harvest; however, none from the open harvest group required supplementation.¹⁰ Three studies did not report any augmentation of graft volume with demineralized bone graft.^{5,12,13} Sharma and colleagues reported that sufficient graft volume could be achieved with trephine harvest, estimating up to 6 cc through three passes with the trephine (estimated 2 cc per pass).⁵

Complications

Although no complications were reported in the trephine group, there were a small number in the open harvest patients: one postoperative infection, one hematoma, one report of paresthesia at donor site, and three cases of painful ambulation. Also reported was one case of idiopathic urinary retention, and one case of seizure activity.

Discussion

Compiled results of the five manuscripts that met our inclusion criteria demonstrated that patients undergoing trephine bone harvest had shorter lengths of stay, shorter operative time,

and lower postoperative narcotic requirements as compared to open bone graft harvest. These results have been similarly demonstrated in adult studies.⁴ However, the volume of cancellous bone harvested by trephine was less than that harvested by the traditional open technique, and often required augmentation with demineralized bone graft. Complications were rare with both techniques but more common in the open harvest group.

Donor Site Morbidity

Although trephine harvest was associated with decreased pain scores and reduced opioid consumption in all included studies, there were significant discrepancies in the use of adjunctive pain control modalities among these studies. Several institutions routinely employed local anesthetic infiltration, intravenous analgesics, regional nerve blocks, anesthetic infusion pumps, or intraoperative placement of collagen sponges (ie, GelFoam®) infiltrated with local anesthetic. As all of these adjuncts are independently associated with reduced postoperative pain, this likely confounds the comparative findings of postop pain reduction, time to ambulation, and length of stay.¹⁴⁻¹⁷

Recent work has assessed the effects of analgesic adjuncts independent of harvest techniques. Shenoy and colleagues reported significantly reduced postoperative pain scores in ICBG patients who received ultrasound-guided transversus abdominis plane (TAP) blocks, as compared to those who received local anesthetic infiltration alone.¹⁷ Patients with a TAP block also had a shorter time to ambulation, which was felt to be due to reduced pain at the donor site.¹⁷ Bykowski et al found that addition of an anesthetic infusion at the donor site significantly reduced length of hospital stay and pain scores on both open and trephine harvest groups, although more so in the latter.¹⁰ These results highlight the importance of pain adjuncts for minimizing postoperative morbidity, regardless of harvest technique.¹⁰

Historically, reported complications of open ICBG harvest have included intraoperative blood loss, dysesthesias, contour deformity, and persistent pain or gait abnormality.^{18,19} Many of these reports were published more than 20 years ago. In both groups reported here, there were very few complications. Contemporary studies have focused on a minimally invasive approach to the open technique, which attempts to limit pain by using a smaller incision, harvesting the medial table, and limiting periosteal stripping. This technique has led to significantly improved outcomes in donor site morbidity.^{1,2,6,20-23}

Volume and Type of Harvested Bone

Mean volumes of cancellous bone obtained with trephine harvest were less than with the open technique and often required supplementation with demineralized bone graft, or multiple passes with the trephine to achieve the desired volume.¹⁰ Although we limited our inclusion criteria to studies comparing open harvest and trephine, other authors have examined the use of either trephine alone or open alone

Table 2. Summary of Included Studies.

Author	Year	Sample size	Demographics	Outcomes	
Bykowski et al ¹⁰	2019	72 (22 open, 14 trephine, 25 trephine with pain pump ^a , 11 no harvest)	Range 9-14 years Open: 17 UL/5 BL ^a , Trephine only: 6 UL/8 BL, Trephine with PP: 18 UL/7 BL No harvest, 4UL/7 BL	Operative time: Length of stay: Analgesia: Bone graft harvest:	Shortest in "Trephine with Pain Pump" group Hospital stay was similar between Open and Trephine-Only. The hospital stay for the "Trephine with Pain Pump" group was lower than a No Harvest comparison group. Almost half of patients from the "Trephine with Pain Pump" group were discharged home directly from PACU. Trephine with Pain Pump group received intravenous acetaminophen, bilateral infraorbital nerve blockade with bupivacaine, and hip donor site ropivacaine infusion catheter. Postoperative oral pain management was similar for all groups, which included acetaminophen and an oral opioid medication Half of patients who underwent trephine harvest were augmented with allograft.
Sharma et al ⁵	2011	104 (55 trephine vs 49 open)	9 years, Range 9-30 years. Open: 25 uni, 24 BL Trephine: 35 UL, 20 BL	Operative time: Length of stay: Analgesia: Bone graft harvest:	Shorter with the trephine compared with open group. Reduced length of stay in trephine group. Trephine group required significantly less postoperative analgesia than open harvest, including narcotic and non-narcotic. Trephine patients had lower pain scores on discharge. No pain adjuncts reported. Tylenol and Codeine provided postoperatively, and opioid PCA as needed. The typical volume of bone harvested with the trephine was 6 cc, with three passes at 2 cc each. Multiple passes of the trephine used to achieve necessary volume. Open volumes not reported, however anecdotally referenced that significantly more bone was harvested in the open technique. No demineralized bone was used to supplement.
Thuaksuban et al ¹¹	2010	21 (11 open vs 10 trephine/mixed with demineralized bone graft)	10.2 ($\pm$ 1.7), range 9-12 years Open: 11 UL, 3 BL Trephine: 10 UL, 3 BL	Operative time: Length of stay: Analgesia: Bone graft harvest:	No significant difference in the operation time between groups. Duration of postoperative hospital stay was reduced in trephine group. Time to unassisted ambulation was significantly less in the trephine group. No significant difference in pain scores. Acetaminophen and meperidine (Demerol), postoperatively. No other pain adjuncts described. Volume of bone graft with open was approximately double the amount harvested with trephine.
Burstein et al ¹²	2000	55 (21 open vs 21 trephine vs 13 bone grinder)	11.2 years (6.5-16 years) Open: 13 UL, 8 BL Trephine 10 UL, 11 BL Bone grinder: 9UL, 4 BL	Operative time: Length of stay: Analgesia: Bone graft quality/harvest:	Total operative time not reported, however, harvest time of iliac crest graft was significantly shorter in the trephine group. Time to ambulation was less in the trephine group, as was length of stay postoperatively. Perioperative analgesia included Local bupivacaine subcutaneous and deep periosteal infiltration. Placed absorbable collagen sponge impregnated with bupivacaine at donor site. Tylenol and codeine postoperatively, with shorter duration of postoperative analgesia requirements in the trephine group than open. Not reported.
McCanny et al ¹³	1998	29 (13 open vs 16 trephine)	14 years, (7-30 years) Open 8 UL, 5 BL, trephine 9 UL, 7 BL	Operative time: Length of stay: Analgesia: Bone graft harvest:	Not reported. No significant difference in hospital stay. Not reported. Not reported.

Demineralized allograft bone graft.

Abbreviations: UL, unilateral; BL, bilateral.

^aRopivacaine infusion catheter.

(without control groups for comparison). In an early description of the use of trephine harvest, McGurk et al suggested that up to 6 cc of cancellous bone could be harvested.²⁴ Missiuna et al reported that 4 to 5 dowels with a 4.5 mm trephine would typically be sufficient for alveolar grafting, and rarely, a second incision was made to harvest additional bone; the volume harvested, however, was not reported in this study.²⁵

Stranix et al assessed the potential volume of cancellous bone graft harvest based on CT measurements.⁸ The maximum cancellous volume they projected that could be safely harvested increased with age, with 12 cc in 8-year-olds, to 17 cc in 12-year-olds, using a 42 mm drill bit.⁸ Despite these predictions, such harvest yields were not corroborated in clinical studies. Caminiti et al performed a study using 4 mm motorized trephine in adult cadavers, as well as 11 live patients.⁷ On average, they achieved 0.46 cc of cancellous bone per core sample in the patient population, for a total of 2.3 to 3.2 cm³ with 5 to 7 passes of the trephine.⁷

Although most studies focus on cancellous bone graft harvest, many surgeons routinely utilize small segments of cortical bone for the reconstruction of the nasal floor defect. The trephine harvest approach limits access to obtaining cortical bone, and thus may be seen as a disadvantage to this technique. The typical trephine size used in pediatric patients would only allow for a very small cortical cap to be harvested, and furthermore, this area is generally cartilaginous in children and thus not suitable as a cortical bone graft.

Insufficient bone graft harvest, and the need for bone substitute have been separately assessed based on CT imaging of alveolar bone height. There is some evidence to suggest that bone substitutes and autogenous bone graft (or a combination) may have similar bone height in the long term, which may obviate the need for a donor site altogether.^{26,27}

Safety

None of the studies included in our analysis reported significant complications with either the trephine or open technique. Stranix and colleagues investigated the theoretic risk of visceral injury using CT imaging, comparing drill size and trajectory.⁸ This group reported no cases of visceral injury in 129 patients who underwent trephine bone graft harvest. They postulated that limiting the trephine depth to 3 cm reduces the risk of visceral injury based on their CT findings.⁸ Caminiti et al corroborate these results, stating an increased risk of fracture to the inner or outer table with trephine depth more than 30 mm.⁷

Missiuna et al describe the use of a Steinman pin as a trochar inserted into the iliac crest, over which they guided the circular trephine during bone graft harvest.²⁵ They state that "occasional breach of the iliac wall" occurred during harvest, and while the frequency of this occurrence was not reported, they did not report any incidence of nerve or visceral injury.²⁵

The current trend in ABG surgery is toward operative intervention in younger age groups, including patients as young as 6 years of age.¹⁵ Computed tomography studies have shown that the cancellous bone width increases significantly from 8 to 12

years of age (7.21 to 10.97 mm, $P < .001$).⁸ Although there have been few reported complications with trephine use in the studies reported here, the safety profile in this cohort with a thin iliac crest is still largely unknown and requires larger studies.

Study Limitations

This systematic review was limited by the small number of studies that met inclusion criteria for direct comparison of open versus trephine harvest technique. Many studies with descriptive techniques, but lacking in comparative analysis, were included in the discussion to further expand on our findings and nurture discussion on the topic. Furthermore, these studies employed a variety of different postoperative pain control strategies making accurate and objective comparison difficult. Small sample sizes may limit the ability to identify rare, but potentially serious, complications such as visceral organ injury.

There may also be some limitation in the applicability of results, as the mean ages reported were 9.0 to 14.3 years. Although debate remains about the ideal time for alveolar grafting, it is typically recommended during the period of mixed dentition, prior eruption of the cleft-side canine.^{15,28}

This review did not assess less commonly used techniques for ICBG harvest, including percutaneous needle, cylindrical osteotome, Volkmann curette, modified bone core biopsy trocar, motorized CORB needle, bone grinder, grinding harvester, or mechanized coring. It is possible that these techniques have utility in specific clinical situations.

Conclusions

Although there are potential advantages to the trephine harvest of ICBG in children, these must be balanced against the significant risk of unpredictable harvest volume and the inability to obtain cortical bone. The limited literature available indicates reduced operative time, time to ambulation, length of hospital stay, and a favorable safety profile for trephine harvest despite the theoretical risk of visceral organ injury in children. Importantly, the evidence would also suggest that employing minimally invasive techniques during open harvest, as well as pain adjuncts such as regional blocks, would abate the historically referenced disadvantages of open harvest. Clinical equipoise still exists regarding the optimal approach for ICBG in the pediatric population and thus will be influenced by operator comfort, preference, and institutional experience.

This systematic review has informed the development of a clinical care pathway for alveolar cleft repair at our institution. We feel that the reported benefits of trephine harvest do not outweigh the disadvantage of limited or unpredictable volumes of cancellous bone and lack of cortical bone access. Consequently, we perform a minimally invasive, short incision open technique for ICBG harvesting, and routinely obtain a section of cortical bone from the inner table for nasal floor reconstruction. Importantly, we have also increased our focus on a standardized

and thorough perioperative pain pathway and employ the use of multiple pain adjuncts, including local and regional anesthetic, and placement of GelFoam® with bupivacaine at the donor site.

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Author Contributions

ART and SF independently conducted the systematic reviews and a priori data extraction, and ART performed analysis with help of a statistician. ART and TRC were largely responsible for manuscript writing, with significant contributions and edits from ARH and FOGF.

Declaration of Conflicting Interests

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Appendix I: MEDLINE Search

1. ((iliac or ilium or bone or cortical or cancellous or alveolar or cleft palate or cleft lip or orofacial cleft* or orofacial cleft*) adj6 (open or osteotome or trapdoor or trap door or conventional or traditional) adj6 (graft* or harvest* or donor* or transplant*))).mp.
2. ((iliac or ilium or bone or cortical or cancellous or alveolar or cleft palate or cleft lip or orofacial cleft* or orofacial cleft*) adj6 (closed or trephine or bur or burr or drill* or motor* or rotary or rotating or piezoelectric or piezo-electric or minimally invasive) adj6 (graft* or harvest* or donor* or transplant*))).mp.
3. 1 and 2